

Geometry
Unit 2
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a picture using the given information.

1. If M is the midpoint of a line segment AB , then $AM = \frac{1}{2}AB$ and $MB = \frac{1}{2}AB$.

Definition

Diagrams may vary.

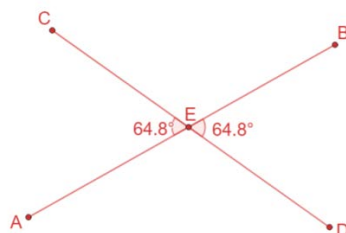


$$AM = \frac{1}{2}AB \text{ and } MB = \frac{1}{2}AB$$

2. Vertical angles are congruent.

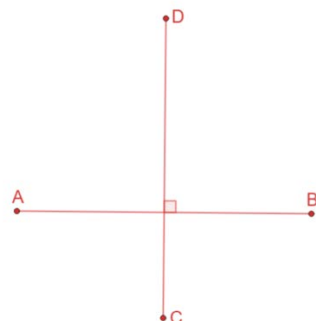
Theorem

Diagrams may vary.



3. Perpendicular lines are lines that intersect to form a right angle.

Definition



Complete the table.

	Property	Equality	Congruence
4.	Transitive	If $m\angle ABC = m\angle DEF$ and $m\angle DEF = 45^\circ$, then $m\angle ABC = 45^\circ$	If $\angle GHI \cong \angle FED$ and $\angle FED \cong \angle ABC$, then $\angle GHI \cong \angle ABC$
5.	Reflexive	$m\angle FGH = m\angle HGF$	$\angle FGH \cong \angle HGF$
6.	Symmetric	If $m\angle XYZ = 19^\circ$, then $19^\circ = m\angle XYZ$	If $\angle XYZ = \angle FGH$, then $\angle FGH = \angle XYZ$

7. Write the conclusion for the statement and name the property.

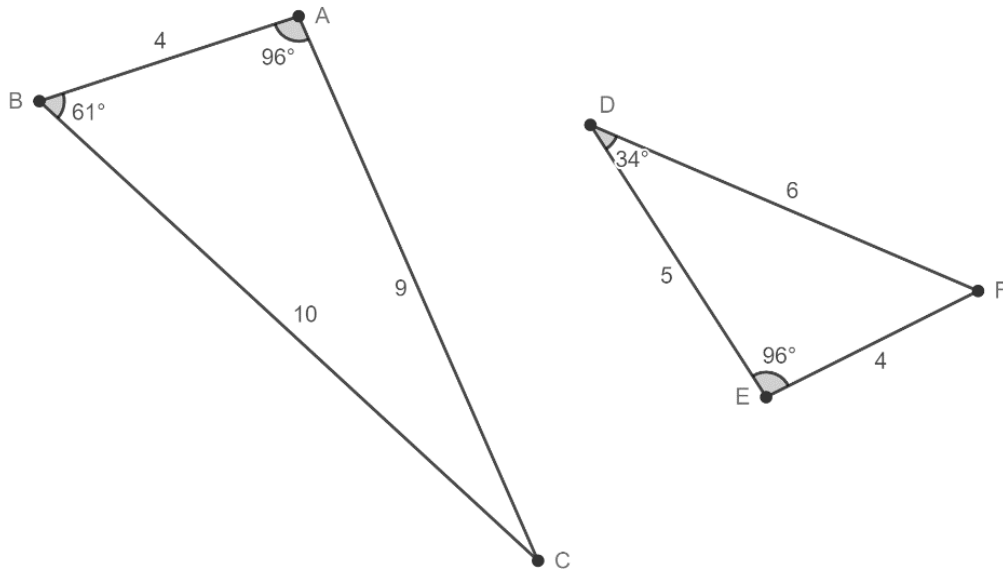
If $m\angle QRS + m\angle LMN = 223^\circ$ and $223^\circ = m\angle GHI$, then...

$m\angle QRS + m\angle LMN = m\angle GHI$; Transitive Property

Complete the table with the reason that makes the given statement true.

	Statement	Reason
8.	$m\angle MNP = m\angle DFG$	Given
9.	$\angle MNP \cong \angle DFG$	Definition of Congruence

10. Triangles ABC and DEF are shown.



Fill in the blanks using the information shown on the two triangles.

$EF = 4$. Therefore, $\overline{EF} \cong \overline{AB}$.

$m\angle CAB = 96^\circ$. Therefore, $\angle CAB \cong \angle DEF$.

Are triangles ABC and DEF congruent? Explain. The triangles are not congruent because corresponding sides are not congruent. (Explanation may vary).